

Recursive Genomic Q-Former: Parameter-Reduction vs. Performance Trade-off

Task details	
Task	cancer_type — 4-way tumor-of-origin classification from bulk gene-expression profiles (TCGA RNA-seq).
Dataset	Unified TCGA cohort: 2,738 samples $\times$ 20,530 genes; top 4,000 highly-variable genes used as input tokens.
Classes	Breast (BRCA, 867), Head-neck (HNSC, 423), Lung (LUNG, 943), Prostate (PRAD, 505).
Split	70/15/15 train/val/test, 523 test samples , seed 42.
Model	Recursive Genomic Q-Former: $d_{\text{model}} = 128$, 256 marker query slots, recursion depth 4.
Metric	Test accuracy and macro-averaged F1.

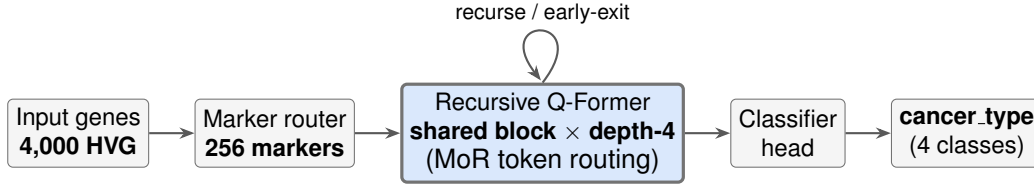


Figure 1: Pipeline. The shared transformer block (blue) is reused across recursion steps; MoR routes tokens to a per-token depth instead of a fixed pass.

All results are on the **cancer_type** classification head (4 classes: BRCA, HNSC, LUNG, PRAD; 523 test samples), with a fixed recursion depth of 4, $d_{\text{model}} = 128$, 256 marker query slots, and seed 42. The “regular” recursive model shares one transformer block across all recursion steps; the comparison models either un-tie those weights (*Independent*) or add token-adaptive routing (*MoR*).

Table 1: Parameters and performance for all three model types. One view of the full trade-off: parameter budget (recurrent-block params, total params, and the independent/shared ratio) alongside all three performance metrics (accuracy, macro-F1, weighted-F1). Sharing the transformer block cuts recurrent params by $\sim 4\times$ for a 0.57-point accuracy cost; Mixture-of-R recursions (MoR) keeps that small budget yet recovers the gap to 0.19 points and is best on macro-F1.

Model	Weights	Routing	Parameters			Performance		
			Recurrent	Total	Ratio	Acc.	Macro-F1	Wtd-F1
Independent stack	untied	none	530,432	1,109,897	$3.99\times$	0.9904	0.9891	0.9904
Recursive (shared)	shared	none	132,992	712,457	$1.00\times$	0.9847	0.9871	0.9847
MoR (token-adaptive)	shared	token	132,992	712,457	$1.00\times$	0.9885	0.9892	0.9885

Read-out. The shared recursive model uses **74.9% fewer** transformer parameters than the independent stack (132,992 vs. 530,432) for a **0.57-point** accuracy cost. Adding MoR token routing

closes that gap to **0.19 points** at the same small parameter budget, and is the only configuration that also reduces FLOPs.

Table 2: Parameter scaling vs. recursion depth. Shared-weight recursion keeps the transformer parameter count essentially flat as depth grows, whereas an independent stack grows linearly. The reduction factor reaches nearly 8 $\times$ at depth 8.

Recursion depth	Shared params	Independent params	Reduction factor
1	132,608	132,608	1.00 $\times$
2	132,736	265,216	2.00 $\times$
4	132,992	530,432	3.99 $\times$
6	133,248	795,648	5.97 $\times$
8	133,504	1,060,864	7.95 $\times$

Table 3: Compute trade-off (depth-4 variants). Early-exit / adaptive recursion lowers the *effective* mean depth and FLOPs per sample relative to a fixed depth-4 pass, at equal parameter count. MoR achieves the lowest compute and the best F1.

Configuration	Transf. params	Mean depth	MFLOPs/sample	Test acc.
Fixed depth-4 (no early exit)	132,480	4.00	268.4	0.9847
Recursive (shared, early-exit)	132,992	2.75	161.5	0.9847
No recursive refine	132,992	2.75	161.5	0.9771
MoR (token-adaptive)	132,992	2.69	159.1	0.9885

Notes. Param ratio in Table 1 is independent/shared transformer parameters. “Transf. params” counts only the recurrent transformer block (the full model incl. embeddings/heads is $\sim 712\text{K}$ for shared variants, $\sim 1.11\text{M}$ for the independent stack). Compute-saving for the early-exit and MoR variants is $\approx 40\%$ of the fixed depth-4 FLOP budget. Token-reduction at the marker stage additionally compresses 4000 input genes to 256 retained markers (93.6% token reduction) at 98.5% accuracy.

Cohort-wise clinical prediction

The model is evaluated on the **unified TCGA benchmark**: 2,738 patient samples, 20,530 genes (top 4,000 highly-variable genes used as input tokens). Tumor-of-origin (`cancer_type`) is modeled *unified* across all four cohorts; the clinical and outcome targets are modeled *cohort-by-cohort* (breast, lung, thyroid, head&neck), so staging and survival are learned without cross-cohort confounding. Because the cohorts are small and class-imbalanced, all targets are evaluated with **stratified 5-fold cross-validation** and reported as **macro-F1 (mean $\pm$ std)** — accuracy on these splits collapses to the majority-class floor and is identical across models, which hides the true difficulty.

Table 4: Cohort-wise macro-F1 of the three model types (5-fold CV). `cancer_type` is modeled *unified* (pooled across all four cohorts, it is the cohort identity); the five clinical / outcome targets are modeled *separately within each cohort* (breast, lung, thyroid, head&neck), for all three model types. Macro-F1 (% , mean $\pm$ std over stratified 5-fold cross-validation) is reported instead of accuracy: on these small, imbalanced cohorts accuracy collapses to the majority-class floor (and is identical across models), whereas macro-F1 exposes the true difficulty. Best per row in **bold**; recurrent-block parameters under each model.

Cohort	Task	Macro-F1 % (mean $\pm$ std, 5-fold CV)		
		Independent 530,432 par.	Recursive (shared) 132,992 par.	MoR 132,992 par.
Unified (all)	<code>cancer_type</code>	97.9 $\pm$ 0.9	96.9 $\pm$ 0.4	96.7 $\pm$ 0.9
Breast (867)	<code>pathologic_stage</code>	26.4 $\pm$ 3.8	25.1 $\pm$ 3.9	26.6 $\pm$ 3.2
	<code>pathologic_T</code>	26.6 $\pm$ 4.4	26.6 $\pm$ 3.7	24.8 $\pm$ 2.3
	<code>pathologic_N</code>	22.8 $\pm$ 1.3	21.2 $\pm$ 2.6	20.7 $\pm$ 4.6
	<code>os_binary</code>	50.2 $\pm$ 5.7	50.4 $\pm$ 3.9	49.3 $\pm$ 5.9
	<code>tumor_status</code>	51.6 $\pm$ 3.4	47.4 $\pm$ 5.0	50.8 $\pm$ 3.0
Lung (943)	<code>pathologic_stage</code>	26.1 $\pm$ 2.5	24.2 $\pm$ 2.9	27.8 $\pm$ 1.6
	<code>pathologic_T</code>	27.3 $\pm$ 2.2	24.7 $\pm$ 1.6	22.4 $\pm$ 3.3
	<code>pathologic_N</code>	22.7 $\pm$ 2.0	24.1 $\pm$ 3.5	24.5 $\pm$ 2.5
	<code>os_binary</code>	46.1 $\pm$ 1.6	46.9 $\pm$ 0.9	46.8 $\pm$ 0.9
	<code>tumor_status</code>	55.0 $\pm$ 2.8	52.6 $\pm$ 5.2	53.1 $\pm$ 3.2
Thyroid (505)	<code>pathologic_stage</code>	25.3 $\pm$ 3.4	24.5 $\pm$ 4.0	24.6 $\pm$ 4.7
	<code>pathologic_T</code>	24.8 $\pm$ 5.9	24.5 $\pm$ 3.3	25.3 $\pm$ 2.4
	<code>pathologic_N</code>	58.0 $\pm$ 3.4	64.8 $\pm$ 2.7	58.9 $\pm$ 4.9
	<code>os_binary</code>	49.2 $\pm$ 2.2	48.7 $\pm$ 0.2	47.2 $\pm$ 3.0
	<code>tumor_status</code>	50.6 $\pm$ 7.3	47.8 $\pm$ 2.1	47.2 $\pm$ 3.2
Head&neck (423)	<code>pathologic_stage</code>	20.4 $\pm$ 4.9	21.8 $\pm$ 3.9	19.2 $\pm$ 2.3
	<code>pathologic_T</code>	20.8 $\pm$ 3.8	20.4 $\pm$ 3.6	19.5 $\pm$ 2.3
	<code>pathologic_N</code>	25.8 $\pm$ 3.8	24.9 $\pm$ 5.5	28.8 $\pm$ 3.9
	<code>os_binary</code>	47.5 $\pm$ 0.2	47.9 $\pm$ 1.6	48.7 $\pm$ 2.3
	<code>tumor_status</code>	50.8 $\pm$ 4.7	51.2 $\pm$ 4.0	50.0 $\pm$ 2.1

Notes. Macro-F1 %, mean $\pm$ std over stratified 5-fold cross-validation (every sample tested once), recurrent-block parameters under each header (total ~ 1.11 M Independent; ~ 712 K shared and MoR). `cancer_type` is the unified four-cohort model (`results_cohorts_cv/*/unified/`); the per-cohort clinical/outcome results are from `results_cohorts_cv/*/(<cohort>)/`. *Reading the table:* only tumor-of-origin (`cancer_type`, $\sim 97\%$) and node status in thyroid (`pathologic_N`, $\sim 59\text{--}65\%$) carry strong signal; the 4-class staging targets hover near the 25% chance level and the binary outcome targets near 50%, confirming that fine-grained clinical staging is not recoverable from bulk expression at these cohort sizes. No single model type dominates, and differences are mostly within one std.

Single-cell cell recognition (genomap benchmark)

For an apple-to-apple comparison with the genomap paper (Islam & Xing, *Nat. Commun.* 2023), we evaluate the same recursive model, in its three variants, on the genomap classification datasets under the **paper’s exact protocol**: precomputed genomap features (5–10% HVG + entropy cartography), the paper’s train/test split (Tabula Muris 70/30; pancreas integration split), Adam + weight decay (lr 0.001, wd 10^{-5} , batch 128, 150 epochs), and **accuracy** as the headline metric. We restrict to the two datasets the genomap paper uses for *classification* — Tabula Muris cell recognition and the five-dataset pancreas integration; the capsule’s other matrices (sea-squirt developmental stages, retinal-bipolar) were used in the paper for trajectory and clustering/visualization, not classification, so they are excluded here.

Table 5: Single-cell recognition vs. the genomap paper. Test accuracy of our three model variants under the genomap paper’s protocol, with the genomap paper’s own reported genoNet accuracy for reference. On pancreas integration our Independent variant matches the paper’s headline 93%. Best per row (among our variants) in **bold**.

Task	Dataset (cells / cls)	Our model (accuracy)			genomap (paper)
		Independent	Shared	MoR	
Cell recognition	Tabula Muris (54,865 / 55)	0.830	0.829	0.826	Suppl. S4
Data integration	Pancreas (14,767 / 15)	0.935	0.920	0.904	0.93

Notes. Accuracy on the paper’s held-out test set (single split, seed 42). The genomap paper reports 93% on the pancreas integration (6% / 21% above Cell-ID / SingleR; Segerstolpe subset 97%); its Tabula Muris accuracy is given only in Supplementary Fig. S4. Our recursive model reaches the paper’s pancreas accuracy at $\sim 4\times$ fewer recurrent parameters (shared / MoR) than the Independent stack (see Table 1). Source: `results_singlecell_paper/*/`.